Correlation matrix and Denoising

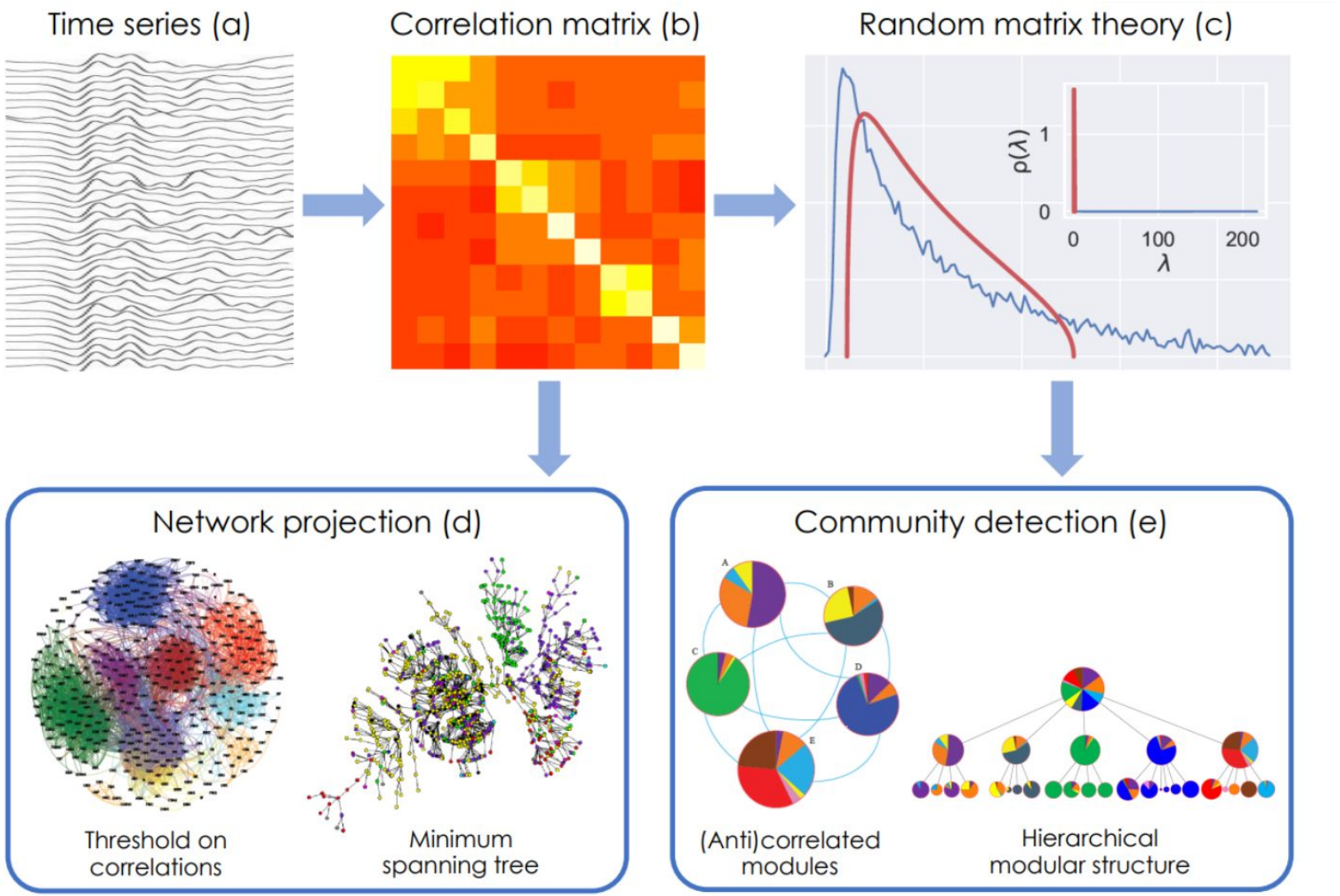
Network edge with correlation matrix

- Appropriate?
- Alternative
- How to care?

Appropriate

1. sparsifying due to non-zero edges
2. not appropriate to apply knife-edge cut for all pair
    - due to heterogeneous entities
    - then which filterings?
        - **minimum spanning trees, maximally planar graph, general manifolds**
3. remove market trend
4. do data have enoug *T*? curse of dimensionality

How to care


Time series (a)
Correlation matrix (b)
Random matrix theory (c)
ρ(λ)
1
0
0
100
200
λ
Network projection (d)
Threshold on correlations
Minimum spanning tree
Community detection (e)
(Anti)correlated modules
Hierarchical modular structure

Random Matrix theory

Marcenko-Pastur theorem

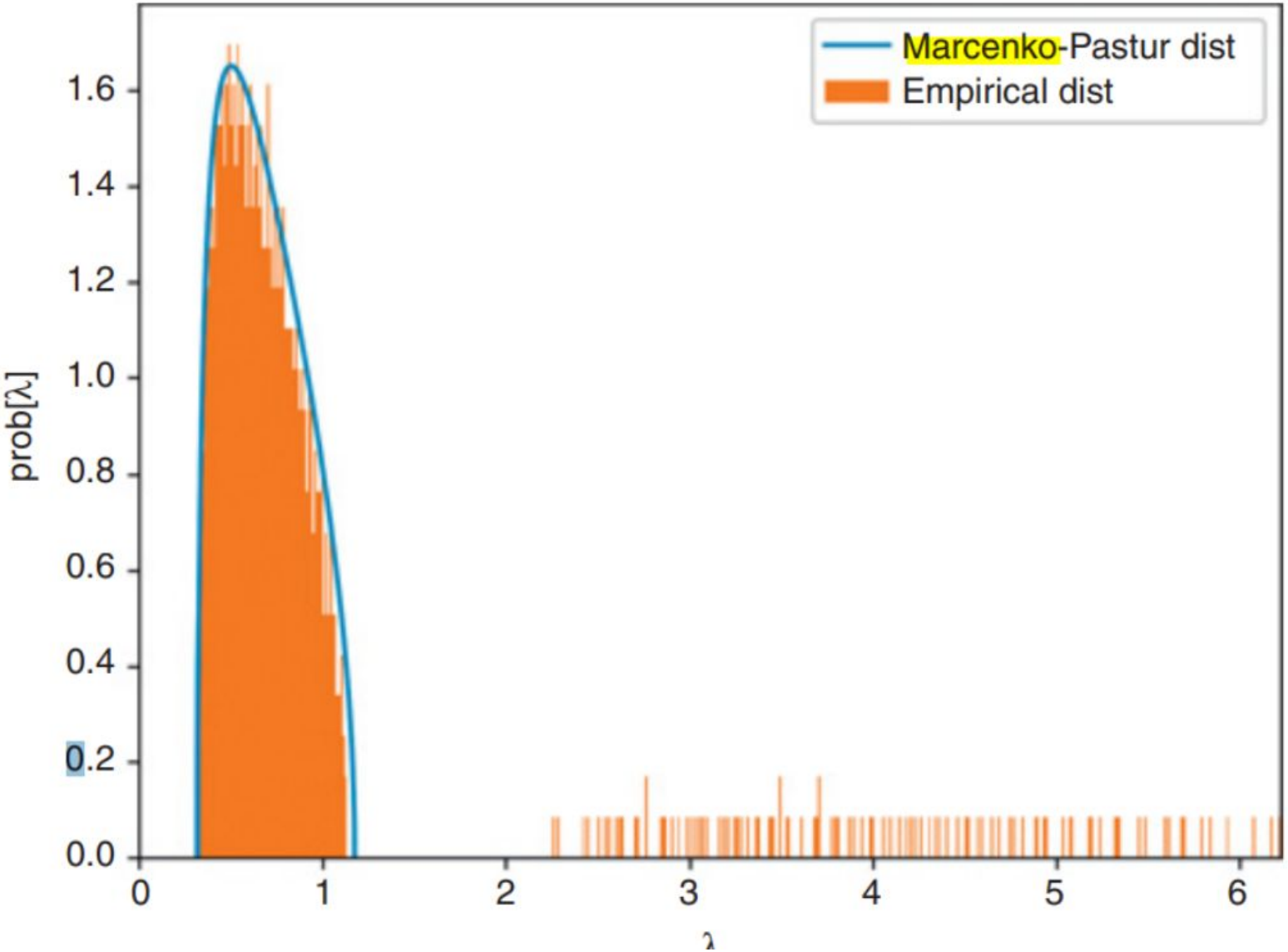
- closed form solution on eigenvalue distribution
- $X : (T \times N) : iid: (0, \sigma^2)$
  $C = X'X/T$
- $\lambda_i's$: eigen value of C

$$f(\lambda) = \begin{cases} \frac{T}{N} \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{2\pi\lambda\sigma^2}, & \text{if} \quad \lambda \in [\lambda_-, \lambda_+] \\ 0 & \text{else} \end{cases}$$

where $\lambda_+ = \sigma^2\left(1 + \sqrt{N/T}\right)^2, \;\; \lambda_- = \sigma^2\left(1 - \sqrt{N/T}\right)^2$

- when $\sigma^2 = 1$, then C is the correlation matrix

Random Matrix theory



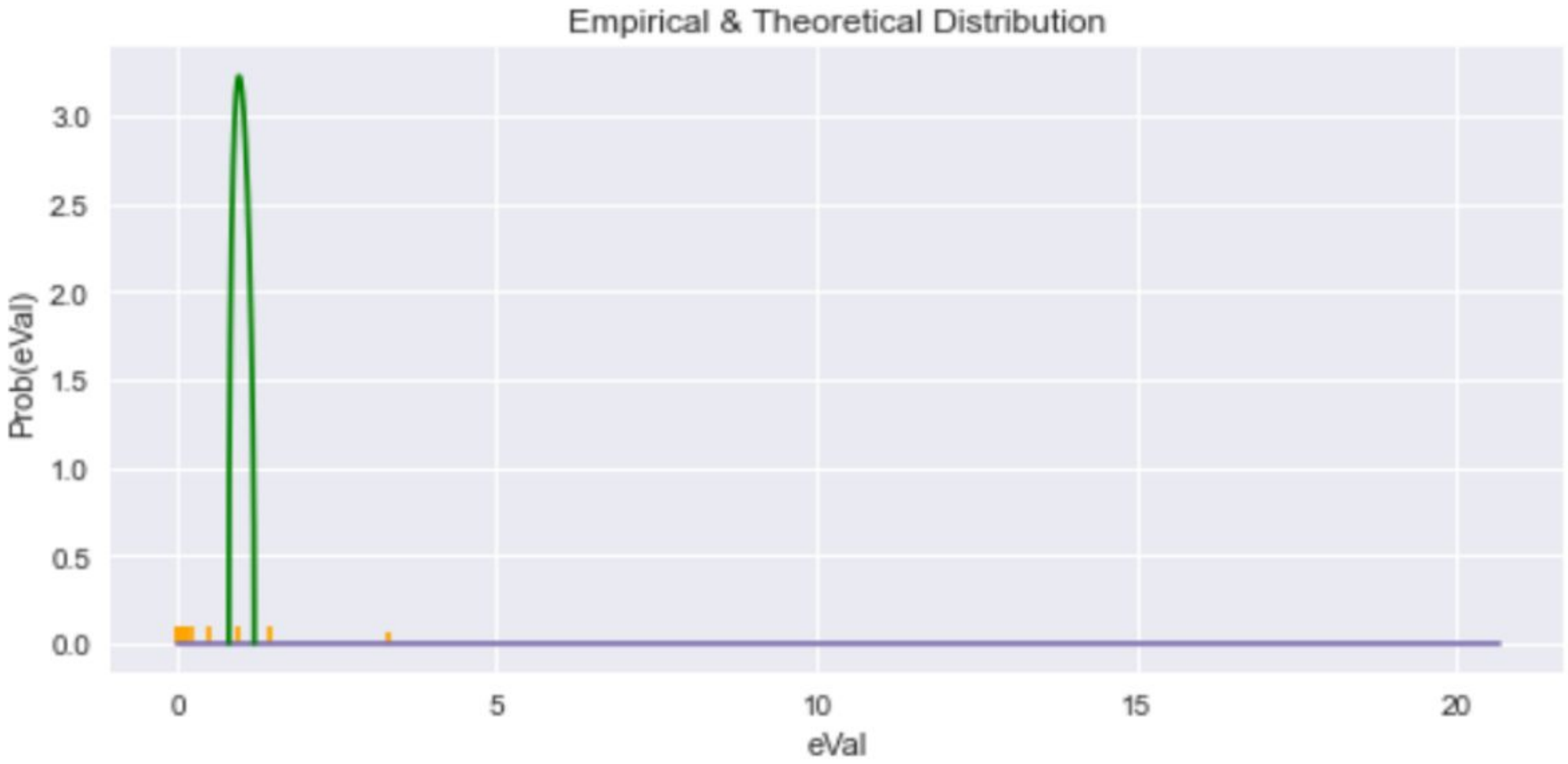
Caveats on MP distribution

1. $\lambda \in [\lambda_-, \lambda_+]$: consistent with random
2. Stock returns' fat-tail property increases the largest eigenvalue and semi-circle law no longer holds.

- Bouchaud and Potters(2009) posit that fat tails can massively increase the maximum eigenvalue in the theoretical limiting spectrum of the random matrix

3. however, mpPDF and Tracy-Widom are good approximation to the return correlation matrix and the largest eigenvalue

Denoising with average



Caveats on MP distribution

Constant residual eigenvalue method

- replace eigenvalue equal to or less than λ_+ with average eigenvalue

$$\tilde{\lambda} \equiv \text{eVal[nFacts:]} = \frac{\text{eVal[nFacts:].sum()}}{\text{eVal.shape[0] - nFacts}}$$

- $\tilde{C}_1 = W'\tilde{\lambda}W$, where W = the original eigenvector
- (denoised correlation matrix) = $C_1 = \tilde{C}_1 \left[\left(\text{diag}[\tilde{C}_1]\right)^{0.5} \left(\text{diag}[\tilde{C}_1]\right)^{0.5'} \right]^{-1}$

detoning

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Recent Literature

- Kendal corr. coeff. portfolio
- See NBER 4930925